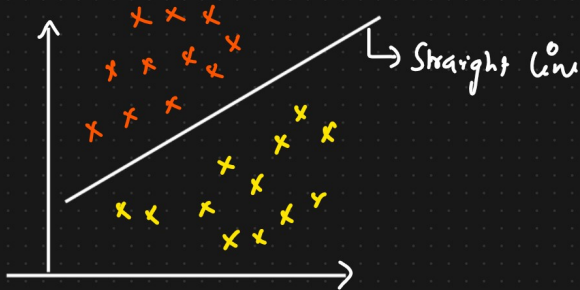
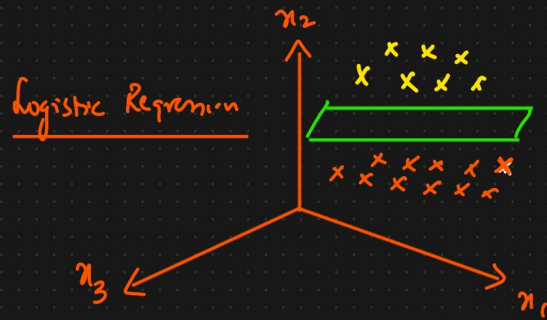


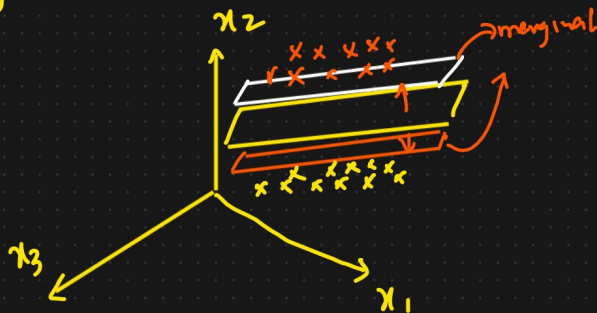
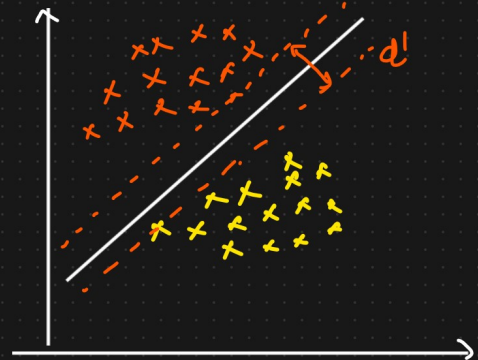
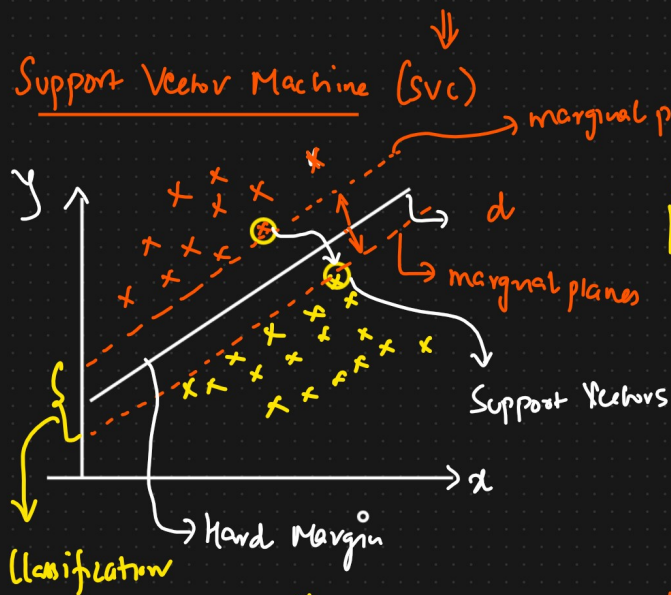
Support Vector Machines ML Algorithms.

① SVC (Support Vector Classifier)

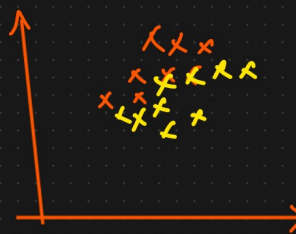
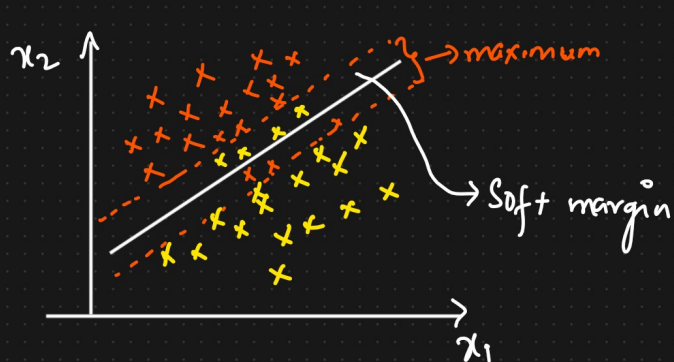
② SVR (Support Vector Regressor)



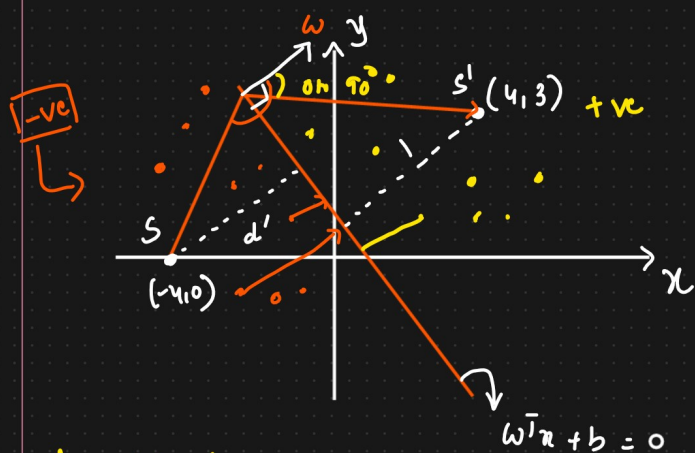
① Support Vector Machine (SVC)



Soft Margin And Hard Margin In SVM



⊛ Support Vector Machines (SVC) Maths Intuition



$$ax + by + c = 0$$

↓

$$w_1 x_1 + w_2 x_2 + b = 0$$

$$\boxed{w^T x + b = 0}$$

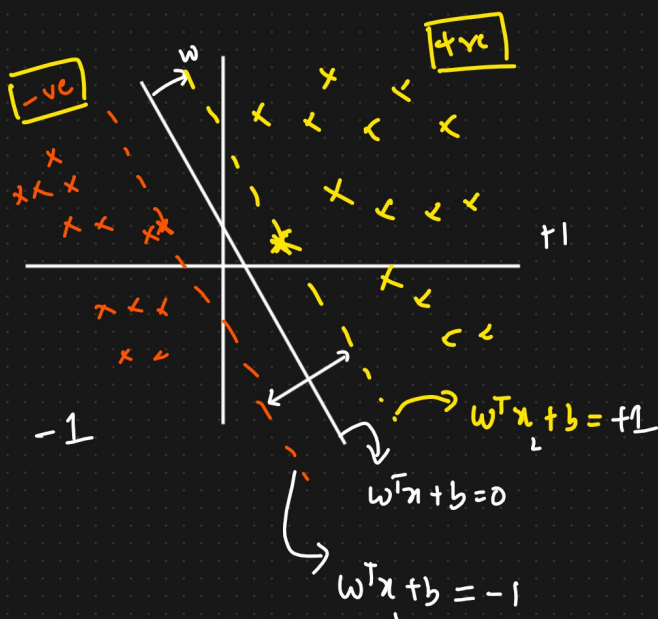
↓

$$b = 0$$

$$\boxed{w^T x = 0}$$

$d = -ve$ below plane

$d = +ve$ above plane



$$w^T x_1 + b = 1$$

$$w^T x_2 + b = -1$$

$$\begin{matrix} (-) & (-) & (+) \end{matrix}$$

$$\frac{w^T (x_1 - x_2)}{\|w\|} = \frac{+2}{\|w\|}$$

Unit vector { Magnitude of the vector is 1 }

Cost function

Maximize $\frac{2}{\|w\|}$ $\Rightarrow$ Distance between marginal plane
 w, b

Constraint such that

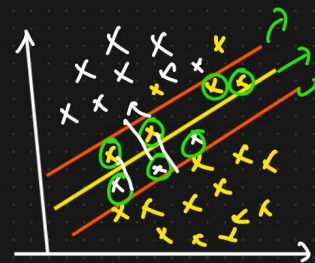
$$y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

For all correct points

Constraint $\rightarrow y_i (w^T x + b) \geq 1$

Maximize $\frac{2}{\|w\|}$ $\Rightarrow$ Min $\frac{\|w\|}{2}$
 w, b

$C_i = 6$ ✓



Cost function of SVM (svc)

Min $\frac{\|w\|}{2}$
 w, b

$\sum_{i=1}^n C_i \xi_i$

Hinge Loss

$$\frac{1}{2}\|w\|^2 + C \sum (\text{Hinge Loss})$$

$\Downarrow$

Soft margin

{ how many points we want to avoid misclassification }

{ distance of the incorrect data points from the marginal plane }

Support Vector Machine tries to find the hyperplane that maximizes the margin while minimizing classification errors. It achieves this by minimizing the objective function $\frac{1}{2}\|w\|^2 + C \sum (\text{Hinge Loss})$, where the first term maximizes the margin and the second term penalizes misclassified or margin-violating points."